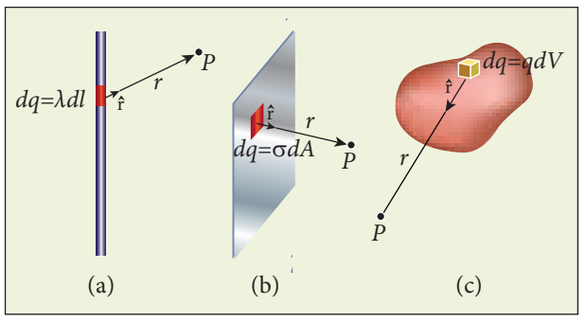


Figure A1.2 Line, surface and volume charge distribution

The electric field due to the line of total charge Q is given by

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\int \frac{\lambda dl}{r^2}\hat{r} = \frac{\lambda}{4\pi\epsilon_0}\int \frac{dl}{r^2}\hat{r}$$

(b) If the charge Q is uniformly distributed on a surface of area A, then surface charge density (charge per unit area) is $\sigma = \frac{Q}{A}$. Its unit is coulomb per square meter (C m^{-2}).

The charge present in the infinitesimal area dA is $dq = \sigma dA$. This is shown in the Figure A1.2 (b).

The electric field due to a of total charge Q is given by

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\int \frac{\sigma dA}{r^2}\hat{r} = \frac{1}{4\pi\epsilon_0}\sigma\int \frac{dA}{r^2}\hat{r}$$

This is shown in Figure A1.2 (b).

(c) If the charge Q is uniformly distributed in a volume V, then volume charge density (charge per unit volume) is given by $\rho = \frac{Q}{V}$. Its unit is coulomb per cubic meter (C m^{-3}).

The charge present in the infinitesimal volume element dV is $dq = \rho dV$. This is shown in Figure A1.2 (c).

The electric field due to a volume of total charge Q is given by

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\int \frac{\rho dV}{r^2}\hat{r} = \frac{1}{4\pi\epsilon_0}\rho\int \frac{dV}{r^2}\hat{r}.$$

A3.1 EXPRESSION FOR TORQUE ON A CURRENT LOOP PLACED IN A UNIFORM MAGNETIC FIELD

Consider a single rectangular loop (this means the number of turns is one) PQRS kept in a uniform magnetic field $\vec{B}$. Let a and b be the length and breadth of the rectangular loop respectively. Let $\hat{n}$ be the unit vector normal to the plane of the current loop which completely describes the orientation of the loop. The direction of magnetic field $\vec{B}$ is shown in Figure A3.1.

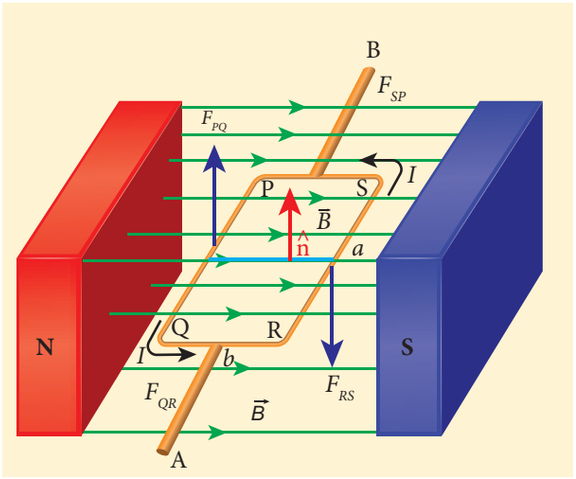


Figure A3.1 Rectangular coil placed in a magnetic field

When a steady current I passes through the loop PQRS, the net force acting on the loop is zero where as the net torque is not zero. For calculation purpose, we shall divide the rectangular loop into four sections PQ, QR, RS and SP. Now we shall consider how

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